

Operator Checklist

◆ BEFORE POWER-ON (GROUND)

Hardware

- ☐ Drone frame intact, props secure
- ☐ Camera mounted firmly (no wobble)
- ☐ Camera angle correct (matches pitch offset)
- ☐ Pixhawk connected to Raspberry Pi
- ☐ GPS antenna unobstructed
- ☐ Battery $\geq 80\%$

Storage & System

- ☐ Raspberry Pi SD card has ≥ 5 GB free
 - ☐ `flights/` directory exists (auto-created is OK)
 - ☐ `best.pt` model present
 - ☐ `config.yaml` verified (no edits needed)
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◆ POWER-ON SEQUENCE

Raspberry Pi

- ☐ Power ON Pi
- ☐ Wait for Pi boot (LED stable / SSH available)
- ☐ Camera recognized (`/dev/video*` exists)

Pixhawk

- ☐ Pixhawk powers ON

- ☐ Heartbeat detected (LEDs normal)
 - ☐ GPS lock achieved (≥ 6 satellites recommended)
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◆ OPTIONAL: LIVE MONITORING (LAPTOP)

If live view required:

On laptop:

```
python tcptagger.py --config config.yaml --show
```

Verify:

- ☐ "TCP connected"
 - ☐ Live video visible
 - ☐ Bounding boxes appear on people
 - ☐ Distance values reasonable
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◆ START SYSTEM (PI SIDE)

Run on Raspberry Pi:

```
python3 tcp_sender.py --laptop-ip <LAPTOP_IP>
```

Verify logs:

- ☐ "Pixhawk heartbeat received"
- ☐ "Recording flight to flights/flight_XXXX"
- ☐ Frames saving in `flights/.../frames/`
- ☐ Log shows `TCP=ON` or `TCP=OFF` (both OK)

📌 Important

- TCP **OFF is acceptable** — system still records
- Do **NOT** stop mission due to TCP OFF

◆ TAKEOFF & FLIGHT

During Flight

- ☐ Maintain altitude ≥ 5 m (better geometry)
- ☐ Avoid extreme pitch ($> 45^\circ$)
- ☐ Avoid sudden aggressive maneuvers
- ☐ Fly slow over target area (≤ 3 m/s)

Detection Behavior

- ☐ One person → **one waypoint**
 - ☐ Revisiting same person → **no duplicate waypoint**
 - ☐ Map markers remain stable (centroid behavior)
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◆ NETWORK FAILURE HANDLING (IMPORTANT)

If:

- Laptop disconnects
- Wi-Fi drops
- TCP switches OFF

👉 **DO NOTHING**

System will:

- Continue saving frames + telemetry
 - Reconnect automatically if possible
 - Allow full offline recovery later
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◆ LANDING & SHUTDOWN

After Landing

- ☐ Disarm drone
- ☐ Stop sender on Pi (**Ctrl+C**)
- ☐ Confirm "Shutdown complete" in logs

Power Down

- ☐ Power OFF Pi safely
 - ☐ Remove battery
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◆ POST-FLIGHT (OFFLINE PROCESSING)

Transfer Data

Copy flight folder from Pi:

```
scp -r flights/flight_XXXX laptop:/path/
```

Run Offline Processing

On laptop:

```
python offline_flight.py \  
  --flight-dir flight_XXXX \  
  --config config.yaml \  
  --out offline_outputs
```

Verify:

- ☐ Annotated images saved
 - ☐ Survivor map generated
 - ☐ Waypoint file created
 - ☐ No duplicate survivor points
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◆ FINAL VALIDATION

- ☐ Open waypoint file in QGroundControl
 - ☐ Waypoints positioned correctly
 - ☐ Altitude reasonable
 - ☐ No absurd far-away points
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DO NOT FLY IF

- GPS shows (0.0, 0.0)
 - Camera feed is black
 - Storage nearly full
 - Props loose
 - Pixhawk heartbeat missing
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SYSTEM STATUS MEANING

Indicator	Meaning
TCP ON	Live streaming active
TCP OFF	Offline-only (SAFE)
Waypoint added	New survivor detected
No new waypoint	Same survivor revisited



OPERATOR CONFIRMATION

If all steps above are satisfied:



SYSTEM CLEARED FOR MISSION
